AI Audio Streaming: Methodology & Cost Analysis

Technical Evaluation of Calling Platform Architectures

Executive Summary

This report provides a comparative study of the primary architectures for implementing real-time conversational AI calling systems. It outlines the implementation methodology and detailed cost structures for four approaches: the Mobile App Gateway, Twilio Media Streams, WebRTC via LiveKit, and Managed Voice APIs (Vapi/Retell).

Approach 1: Mobile App Gateway (Current Architecture)

Methodology:

- **App Transport:** The mobile device acts as an audio transceiver. Dart code establishes WebSocket live loops and requests native phone call routing via MethodChannels.
- **Mic Stream:** Native Kotlin handles AudioRecord mono microphone capturing (16kHz PCM), chunking it into byte arrays sent over WebSockets to the Python backend.
- **Audio Playback:** Incoming ElevenLabs audio bytes are downloaded on the mobile client, written to a temporary MP3 file, and played in real time using the Android MediaPlayer.
- **Transcription & AI:** Python server pipes PCM bytes into Deepgram SDK's live stream. The transcription result triggers Gemini LLM generation, which synthesizes audio via ElevenLabs TTS.

Cost Breakdown:

- **Deepgram STT:** ~$0.0043 to $0.0150 per minute (depending on Tier/Volume).
- **Gemini LLM:** ~$0.00015 per 1K input tokens + $0.0006 per 1K output tokens (extremely low).
- **ElevenLabs TTS:** ~$0.15 to $0.24 per 1,000 characters (avg. ~$0.08 per conversational minute).
- **Telephony (SIM):** Cost of the mobile carrier plan (usually flat-rate or standard cellular billing).
- **Total Estimated Cost: ~$0.10 to $0.15 per conversation minute**.

Approach 2: Twilio Media Streams (Direct Telephony)

Methodology:

- **SIP Integration:** A phone call is initiated via Twilio API. Twilio's XML instruction (`<Stream>`) directs it to connect bi-directionally to the backend's WebSocket url.

• **Audio Packets:** Twilio streams raw audio payload (8kHz Mu-law PCM) formatted in JSON envelopes containing base64 encoded audio strings.

• **Speech Translation:** Python backend decodes Mu-law bytes, upsamples them to 16kHz mono, and routes them to Deepgram.

• **Voice Generation:** Gemini output is sent to ElevenLabs. The resulting audio is downsampled to 8kHz Mu-law PCM, base64 encoded, wrapped in a Twilio JSON envelope, and sent back.

Cost Breakdown:

• **Twilio Voice:** $0.0130 to $0.0140 per minute (Standard SIP/PSTN inbound/outbound rates).

• **Deepgram STT + Gemini LLM + ElevenLabs TTS:** Same as Approach 1 (approx. ~$0.09 per minute).

• **Total Estimated Cost: ~$0.11 to $0.17 per conversation minute**.

Approach 3: WebRTC via LiveKit (App/Web Native Agent)

Methodology:

• **UDP Media Server:** Set up an open-source LiveKit Media Server (SFU) running in Docker. The client app connects to LiveKit Room using WebRTC.

• **Media Pipeline:** LiveKit handles high-quality bi-directional audio tracking using UDP/SRTP protocols (extremely resilient to network drops, sub-100ms transport latency).

• **Agent Integration:** LiveKit Agents Framework running in Python acts as a participant in the room, intercepting client voice tracks and piping synthesized responses directly back as a track.

Cost Breakdown:

• **LiveKit Hosting:** Free (Self-hosted on AWS EC2/DigitalOcean) or LiveKit Cloud (Free tier, then $0.0035/minute).

• **Deepgram STT + Gemini LLM + ElevenLabs:** Same as Approach 1 (approx. ~$0.09 per minute).

• **Total Estimated Cost: ~$0.09 to $0.11 per conversation minute**.

Approach 4: Managed Voice APIs (Vapi / Retell AI)

Methodology:

• **One-API Setup:** No custom WebSockets, audio buffering, resamplings, or SDK synchronization is written.

• **Configuration:** An agent config is created via API specifying: System Prompt, Deepgram Model, ElevenLabs Voice ID, and Webhook URLs.

• **Execution:** Call Vapi/Retell API to dial. Their infrastructure coordinates the entire STT-LLM-TTS stream, providing highly optimized sub-second conversational latency.

Cost Breakdown:

• **Platform Fee:** $0.15 per conversation minute (includes platform overhead + built-in standard STT/TTS pricing).

• **Premium Voice Add-on (ElevenLabs):** Custom ElevenLabs key usage billed at cost (~$0.08 per minute).

• **Total Estimated Cost: ~$0.20 to $0.25 per conversation minute**.

Cost & Architecture Comparison Table

Architecture Approach	Telephony / Media Cost	AI Services (STT/LLM/TTS)	Total Est. Cost/Min	Development Complexity
1. Mobile App Gateway	SIM Carrier Cost (Flat-rate)	~$0.09 / min	**~$0.10 - $0.15**	Medium (Requires Native OS Kotlin/Swift bridges)
2. Twilio Media Streams	~$0.0135 / min (Twilio SIP)	~$0.09 / min	**~$0.11 - $0.17**	High (Requires complex socket payload encoding)
3. WebRTC via LiveKit	Self-hosted ($0) or Cloud ($0.0035)	~$0.09 / min	**~$0.09 - $0.11**	High (Requires WebRTC media server setup)
4. Managed Voice APIs	Included in fee	Platform fee + custom TTS (~$0.23)	**~$0.20 - $0.25**	Low (Single HTTP call config setup)